

# STEVEN ASANTE-POKU JNR

Machine Learning & Cybersecurity Engineer | Cloud & AI Solutions Developer

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## PROFESSIONAL SUMMARY

Machine Learning and Cybersecurity focused Information Technology graduate, certified as an AWS Certified Cloud Practitioner and a Google Certified Cybersecurity Professional, with hands-on experience across the AI and cloud stack. Designed, built, and deployed two production grade AWS fintech systems, OmniGuard MoMo AML and OmniSync MoMo Reconciliation Engine, alongside MediChain, a hackathon-winning cloud health records platform. Combines applied machine learning research, hands-on AWS service design, and cybersecurity fundamentals with strong project management, negotiation, and public speaking skills. Well suited for roles in Machine Learning Engineering, Cybersecurity, Cloud and AI Development, or Data Science.

## EDUCATION

### Bachelor of Science in Information Technology

Pentecost University, Accra, Ghana | January 2023 - June 2026

### Fundamental Theories and Implementations of Data Science

The University of Tokyo | August 2026

## TECHNICAL SKILLS

**Machine Learning & Data Science:** Python, Pandas, NumPy, Scikit-learn, Classification and Regression, Anomaly Detection, Feature Engineering, Model Evaluation

**Cybersecurity:** Risk Assessment, Threat and APT Analysis, Dark Web Operations, NIST Framework, PII Redaction (Amazon Macie), Secure System Design

**Cloud Computing (AWS):** Lambda, API Gateway, DynamoDB, S3, Cognito, SageMaker, Neptune, EventBridge, CloudFront, Amplify, Entity Resolution, Glue, Athena, SAM (IaC)

**AI & Research:** Retrieval-Augmented Generation (RAG), Knowledge Retrieval Systems, Prompt Engineering, Explainable AI (XAI)

**Web & Systems:** JavaScript, React, HTML, CSS, Java, SQL, MySQL, Git, VS Code, CI/CD (GitHub Actions)

**Other Tools:** Canva (Graphics Design), Technical Documentation

## FEATURED PROJECTS

### OmniGuard MoMo AML | Independent Cloud/AI Project

- **Built a serverless fraud detection pipeline** using Amazon SageMaker, Neptune graph analytics, and AWS Entity Resolution to detect mule rings and smurfing in real time.
- **Automated 93.5% of alert triage** through a 3-tier confidence engine, cutting analyst workload from 2,000 daily alerts to about 130 gray-zone cases.
- **Delivered sub-200ms scoring with explainable AI narratives** and a React fraud-analyst dashboard, secured with a full defense-in-depth AWS architecture.

Repository: [github.com/Steven256-debug/omniguard-momo-aml](https://github.com/Steven256-debug/omniguard-momo-aml)

### OmniSync MoMo Reconciliation Engine | Independent Cloud Project

- **Built an event-driven serverless architecture** (API Gateway, Lambda, SQS, DynamoDB) to reconcile mobile money payments in real time across MTN, Telecel, and AT.
- **Replaced manual, spreadsheet-based reconciliation** with a unified live ledger, HMAC-secured webhooks, and Cognito-authenticated access.
- **Shipped a production React dashboard** with analytics via AWS Glue and Athena, deployed on Amplify with automated CI/CD.

Repository: [github.com/Steven256-debug/omnisync-momo-reconciliation](https://github.com/Steven256-debug/omnisync-momo-reconciliation)

### MediChain | AWS GenAI Hackathon, 3rd Place

- **Co-built an AI-powered patient health records platform** in a team of 3, handling the full AWS backend (Cognito, DynamoDB, Lambda, Bedrock, Amplify).

## RESEARCH & PROFESSIONAL EXPERIENCE

Research Assistant, AI & Knowledge Systems | Institutional Research Project

- Contributed to RAG-based institutional knowledge systems, improving response accuracy through data structuring and retrieval optimization.

#### **AI Developer, Academic Performance Analytics** | University Project

- Developed AI-powered academic analytics tools, applying ML to identify trends and academic risk indicators.

#### **IT Helpdesk Support Technician** | Amansie West Rural Bank PLC

Jun 2023 - Sep 2023

- Delivered Tier-1 IT support and maintained enterprise systems, improving documentation and ticket workflows.

#### **Cybersecurity Virtual Intern** | DATACOM

Nov 2023 - Jan 2025

- Conducted simulated risk assessments and APT breach analysis, aligning mitigation strategies with NIST standards.

### **MACHINE LEARNING PROJECTS**

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- **Credit Card Fraud Detection:** classification model detecting fraud on a highly imbalanced dataset.
- **Diabetes Classification:** medical ML model predicting diabetes risk from patient health data.
- **California Housing Price Prediction:** regression model predicting housing prices from demographic and geographic data.

### **ACHIEVEMENTS**

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- 3rd Place, AWS Cloud Computing / GenAI Hackathon (MediChain)

### **CERTIFICATIONS**

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- AWS Certified Cloud Practitioner, Amazon Web Services Training and Certification (issued August 2026)
- Google Certified Cybersecurity Professional, Google
- Introduction to Dark Web Operations, Security Blue Team (November 2024)
- Introduction to Cybersecurity, Cisco Networking Academy (November 2024)
- Cybersecurity Virtual Internship, APT Breach Analysis and Risk Assessment, DATACOM

### **CORE COMPETENCIES**

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Machine Learning Model Development • Applied AI Research • Cloud Solutions Development • Cybersecurity Risk Assessment • Data Analysis • IT Project Management • Negotiation • Public Speaking • Technical Documentation • Detail-Oriented • Adaptable • Effective Communication